Table 1: Leave-one-out ablation of the two core contributions on CORE4D-G1 (all 50 paper sequences). Each ablation removes one sub-component and re-runs the sampling-based MPC; **Ours** is the full method. *Contribution 1 (Multi-Scale Contact)*: −SB removes the fine SDF surface-band term, −HDMI removes the coarse anchor-attraction term. *Contribution 2 (Part-wise Penetration)*: −gate removes all hard candidate gates, −pen removes all soft penetration penalties. MPJPE is the mean per-joint Cartesian error over all robot links (whole-body, not just root/eef): MPJPE-G is global, MPJPE-L is pelvis-aligned (pose only). ↑/↓: higher/lower is better. Note −HDMI's lower penetration is an artifact of failing to reach the object (see MPJPE-G), not better contact.

Metric	Ours	w/o local reward	w/o global reward	w/o hard select	w/o soft penalty
Phys. contact (%) ↑	**72.5**	39.6	47.2	70.0	<u>71.6</u>
Phys. penetration (%) ↓	**7.6**	12.6	<u>7.9</u>	8.7	8.5
Max phys penetration (mm) ↓	<u>12.0</u>	16.1	**11.6**	13.1	12.8
MPJPE (cm) ↓	<u>12.34</u>	**12.28**	40.07	18.88	15.78
Obj pos err (cm) ↓	<u>10.24</u>	**9.74**	10.34	10.74	10.48
Obj ori err (deg) ↓	5.01	<u>4.99</u>	**4.97**	5.47	5.60